Olist E-Commerce Analytics Dashboard

Newton School of Technology •
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GITHUB | TABLEAU DASHBOARD

Executive Summary

Core Finding

Delivery delays are the **most critical factor** impacting customer ratings. On-time orders consistently receive higher satisfaction scores.

Geographic Risk

Revenue is heavily concentrated in **São Paulo**, indicating a lack of geographic diversification and increased business risk.

Payment Dependency

Credit card dominance introduces **financial dependency risk**, limiting flexibility and resilience.

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Sector Context & Problem Statement

The Challenge

Rapid e-commerce growth is accompanied by **inefficient logistics, high competition,** and evolving customer expectations. The primary problem is the imbalance between growing sales and declining satisfaction due to delivery inefficiencies.

Objectives

- Identify key revenue-driving regions & categories
- Analyze delivery performance and delays
- Evaluate customer satisfaction via review scores
- Provide actionable business recommendations

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Data Description & ETL Pipeline

Dataset Overview

The Olist dataset covers **100,000+ transactions** (2016–2018) across relational tables: orders, customers, sellers, products, payments, and reviews.

- Order timestamps, delivery dates, payment types
- Product categories and customer ratings
- Limitations: missing delivery values, Brazil regional bias, no real-time updates

ETL Pipeline (Python)

Missing Values

Imputation and removal techniques applied

Feature Engineering

Delivery time and late delivery flags created

Outlier Removal

Payment value outliers removed for consistency

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KPI Framework

Five key performance indicators provide a comprehensive view of business performance, operational efficiency, and customer satisfaction — each directly mapped to strategic objectives.

Revenue

Total order value tracked by region and category

Order Volume

Monthly transaction counts and trends

Avg Delivery Time

Days from order to fulfillment

Customer Rating

Review scores 0–5 across all orders

Late Delivery Rate

Percentage of orders delivered past承诺 date

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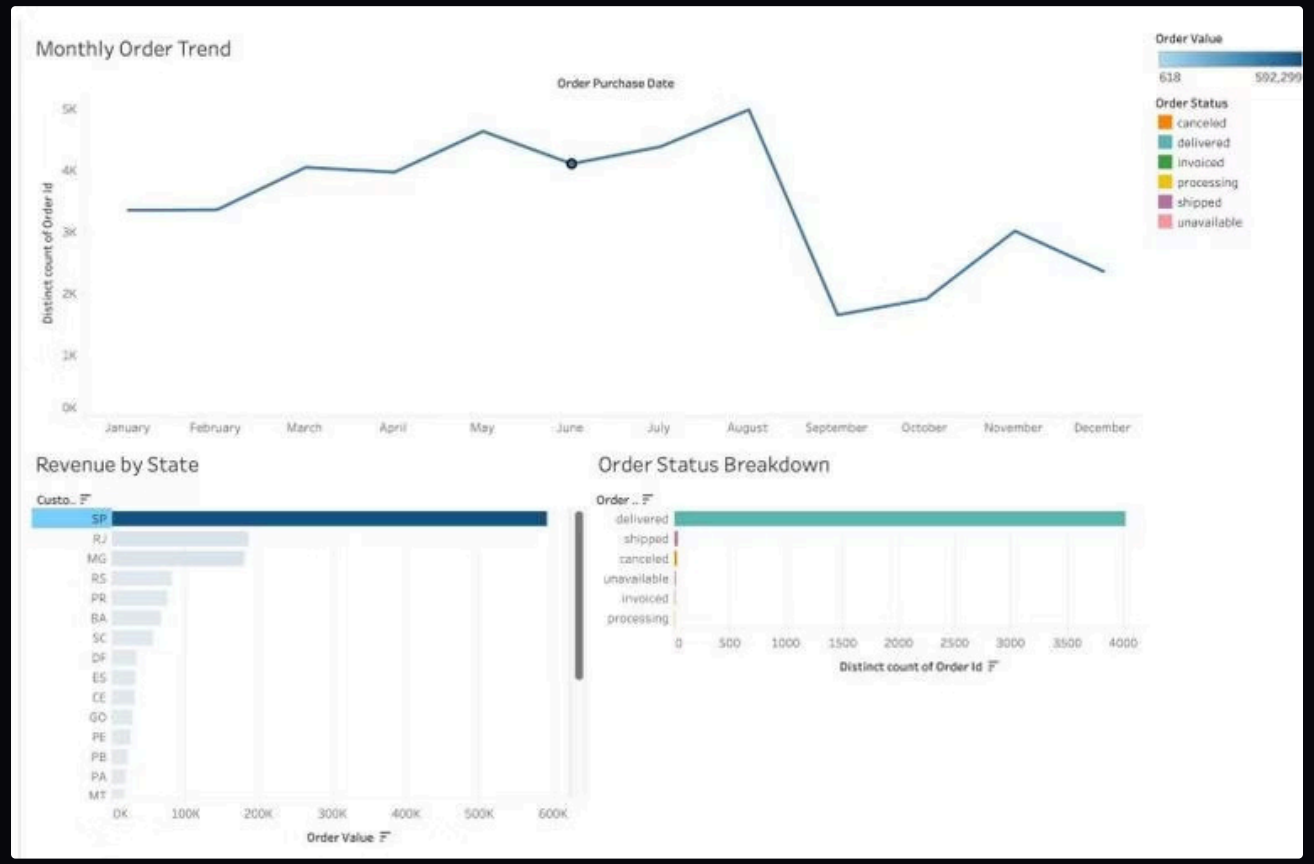
Dashboard Analysis — Operations



Key Observations

- **Monthly Orders:** Peak of ~4.8K in August, sharp decline to 1.8K in September, recovery to ~3K in November
- **Revenue by State:** SF leads at ~600K, far ahead of other states
- **Order Status:** ~3,800 delivered; cancellations and unavailable orders reveal operational inefficiencies
- **Payment Risk:** Credit cards dominate, creating dependency
- **Logistics Impact:** States with longer delivery times show significantly lower ratings

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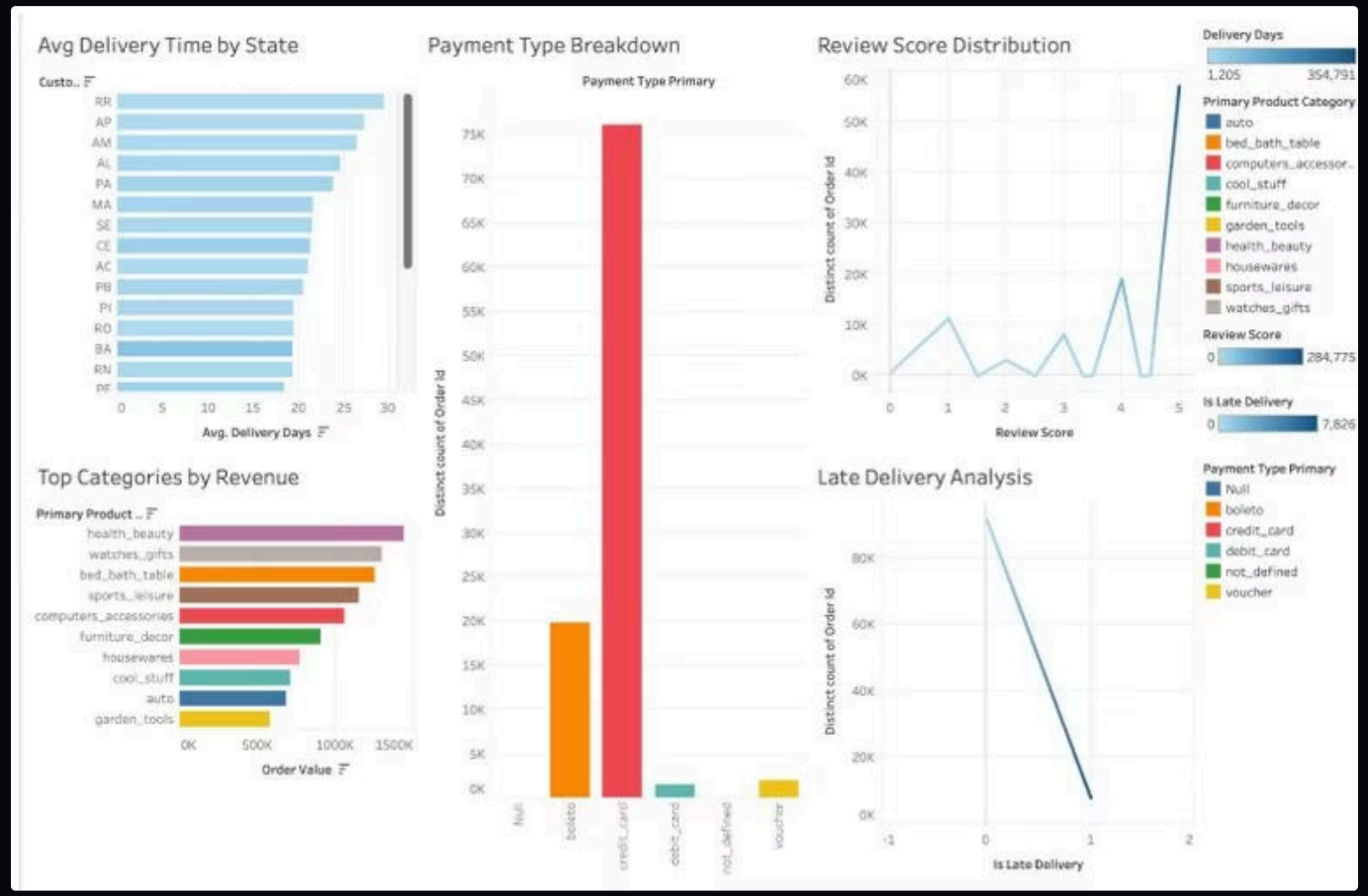
Dashboard Analysis — Revenue & Trends



Five-Chart Insights

- **Delivery Time:** São Paulo averages ~28 days — highest among states
- **Payment Types:** Credit cards account for 75,000+ orders
- **Review Scores:** Strong peak at score 5, but delays drive negative feedback
- **Top Category:** Health & Beauty leads at ~150K BRL revenue
- **Late Deliveries:** Sharp decline over time, but impact on satisfaction remains critical

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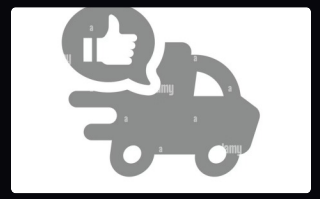
EDA & Statistical Analysis

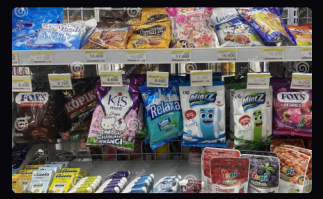


Delivery Time vs. Ratings

EDA reveals a **strong negative correlation** between delivery time and customer ratings. Faster deliveries consistently yield higher scores; delays significantly reduce satisfaction and repeat purchases.

Geographic Concentration Risk

Revenue concentration in a few states increases business risk. High-value customers cluster in **urban areas** — improving service quality there can significantly boost revenue.



Category Demand Imbalance

Category-wise analysis shows **uneven demand distribution**, indicating the need for diversification strategies to ensure sustainable, resilient growth.

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Insights & Recommendations

1

Optimize Logistics

Improve infrastructure in high-delay regions to directly boost customer satisfaction and ratings.

2

Diversify Payments

Reduce credit card dependency by expanding alternative payment methods for financial resilience.

3

Expand Geographically

Enter underperforming regions to reduce revenue concentration and unlock new market potential.

4

Demand Forecasting

Implement seasonal forecasting to manage fluctuations and improve inventory and delivery planning.

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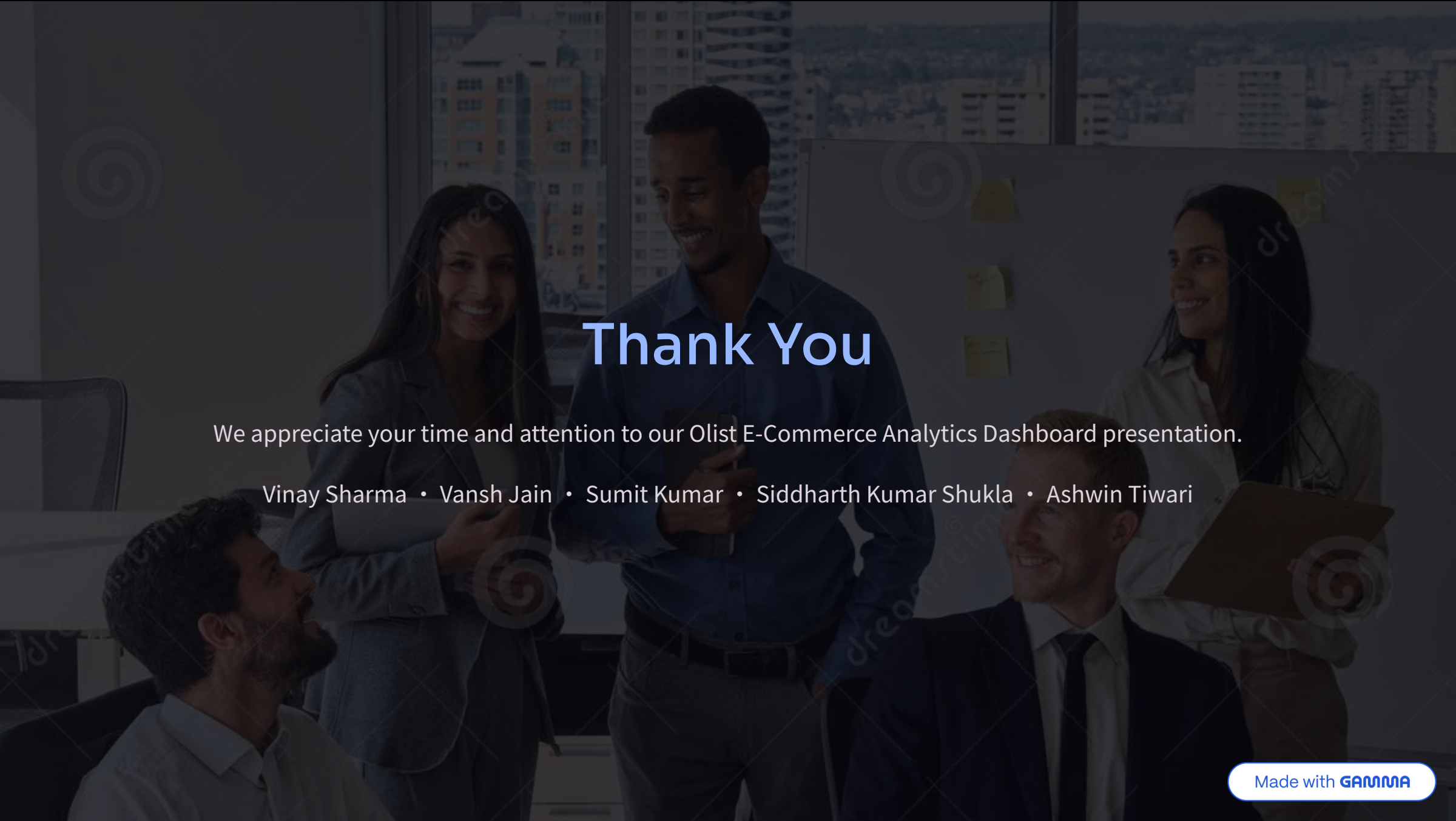

Thank You

We appreciate your time and attention to our Olist E-Commerce Analytics Dashboard presentation.

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